

Deduplicating Training Data Makes Language Models Better

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Abstract

We find that existing language modeling datasets contain many near-duplicate examples and long repetitive substrings. As a result, over 1% of the unprompted output of language models trained on these datasets is copied verbatim from the training data. We develop two tools that allow us to deduplicate training datasets—for example removing from C4 a single 61 word English sentence that is repeated over 60,000 times. Deduplication allows us to train models that emit memorized text ten times less frequently and require fewer training steps to achieve the same or better accuracy. We can also reduce train-test overlap, which affects over 4% of the validation set of standard datasets, thus allowing for more accurate evaluation. Code for deduplication is released at <https://github.com/google-research/deduplicate-text-datasets>.

1 Introduction

A key factor behind the recent progress in natural language processing is the development of large-scale text corpora used to train increasingly large language models. These datasets have grown from single gigabytes to as much as a terabyte over the past few years (Chelba et al., 2013; Xue et al., 2020; Graff et al., 2003; Brown et al., 2020). Because it is so expensive to perform manual review and curation on massive datasets, they tend to suffer in quality compared to their smaller predecessors. This has implications far beyond metrics like perplexity and validation loss, as learned models reflect the biases present in their training data (Bender et al., 2021; Wallace et al., 2019; Sheng et al., 2020). Quantitatively and qualitatively understanding these datasets is therefore a research challenge in its own right (Dodge et al., 2021a).

We show that one particular source of bias, duplicated training examples, is pervasive: all four common NLP datasets we studied contained duplicates. Additionally, all four corresponding validation sets contained text duplicated in the training set. While naive deduplication is straightforward (and the datasets we consider already perform some naive form of deduplication), performing thorough deduplication at scale is both computationally challenging and requires sophisticated techniques.

We propose two scalable techniques to detect and remove duplicated training data. *Exact* substring matching identifies verbatim strings that are repeated. This allows us to identify cases where only part of a training example is duplicated (§4.1). *Approximate* full document matching uses hash-based techniques (Broder, 1997) to identify pairs of documents with high n -gram overlap (§4.2).

We identify four distinct advantages to training on datasets that have been thoroughly deduplicated.

1. Over 1% of tokens emitted unprompted from a model trained on standard datasets (e.g., C4) are part of a memorized sequence (See §6.2)—even though the 1.5 billion parameter model is much smaller than the 350GB dataset it was trained on. By deduplicating the training dataset we reduce the rate of emitting memorized training data by a factor of $10\times$.
2. Train-test overlap is common in non-deduplicated datasets. For example, we find a *61-word sequence*¹ in C4 (Raffel et al., 2020) that is repeated 61,036 times verbatim in the training dataset and 61 times in the validation set (0.02% of the samples in each dataset).

¹“by combining fantastic ideas, interesting arrangements, and follow the current trends in the field of that make you more inspired and give artistic touches. We’d be honored if you can apply some or all of these design in your wedding. believe me, brilliant ideas would be perfect if it can be applied in real and make the people around you amazed!”

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This train-test set overlap not only causes researchers to over-estimate model accuracy, but also biases model selection towards models and hyperparameters that intentionally overfit their training datasets.

3. Training models on deduplicated datasets is more efficient. Processing a dataset with our framework requires a CPU-only linear-time algorithm. And so because these datasets are up to 19% smaller, even including the deduplication runtime itself, training on deduplicated datasets directly reduces the training cost in terms of time, dollar, and the environment (Bender et al., 2021; Strubell et al., 2019; Patterson et al., 2021).
4. Deduplicating training data does not hurt perplexity: models trained on deduplicated datasets have no worse perplexity compared to baseline models trained on the original datasets. In some cases deduplication reduces perplexity by up to 10%. Further, because recent LMs are typically limited to training for just a few epochs (Radford et al., 2019; Raffel et al., 2020), by training on higher quality data the models can reach higher accuracy faster.

To summarize, data duplication offers significant advantages and no observed disadvantages. In the remainder of this paper we present our text deduplication framework in §4, and study the extent of duplicate content in common NLP datasets (e.g., C4, Wiki-40B, and LM1B) in §5. We then examine the impact of deduplication on test perplexity (§6.1) and on the frequency of emitting memorized content (§6.2). Finally, we analyze to what extent perplexity on existing, released models are skewed as a result of overlap between the train and test/validation splits (§6.3).

2 Related Work

Large language model datasets. While we believe our results are independent of model architecture, we perform our analysis on Transformer-based decoder-only language models (Vaswani et al., 2017) trained for open-ended text generation. These current state-of-the-art models are trained on internet text. For example, the GPT-2 family of models Radford et al. (2019) is trained on WebText, a dataset of web documents highly ranked on Reddit—however this dataset was not made available publicly. A common dataset starting point

is CommonCrawl, an index of public webpages. Among the models trained on CommonCrawl include GPT-3 (Brown et al., 2020) with the addition of book datasets, GROVER (Zellers et al., 2019) on a restricted subset filtered to news domains called RealNews, and T5 (Raffel et al., 2020) on a cleaned version of common crawl called C4. Other models are trained on more curated Internet sources—for example Guo et al. (2020) used high quality processed Wikipedia text from 40 different languages to train monolingual 141.4M parameter language models. Non-English models necessarily use different datasets; Zeng et al. (2021) for instance introduced PANGU- α , a family of models with up to 200B parameters that were trained on a non-public corpus of cleaned and filtered Chinese-language documents from CommonCrawl and other sources. Since many of these datasets are not public, we deduplicate three that are: Wiki-40B, C4, and RealNews—as well as the One Billion Word Language Model Benchmark (Chelba et al., 2013), a smaller dataset commonly used for evaluation.

Contamination of downstream tasks. When models are trained on datasets constructed by crawling the Internet, it is possible the model will train on the test set of downstream target tasks. For example, Radford et al. (2019, §4) performed a post-hoc analysis to identify 8-gram overlaps between GPT-2’s training set and datasets used for evaluation, and Dodge et al. (2021b) analyzed C4 and found that up to 14.4% of test examples for various standard tasks were found verbatim (normalizing for capitalization and punctuation) in the dataset. A more proactive approach removes contaminated data. Trinh and Le (2018, Appendix B) removed documents from their CommonCrawl-based train set that overlapped substantially with the common-sense reasoning used for evaluation. And GPT-3 (Brown et al., 2020, §5) did the reverse and removed downstream evaluation examples from their training data by conservatively filtering out any train set examples with a 13-gram overlap with any evaluation example. Up to 90% of tasks were flagged as potentially contaminated.

In our research, we do not focus on the impact of duplicate text in pretrained models on downstream benchmark tasks; instead we address how duplicate text in the LM training and validation sets impacts model perplexity and the extent to which generated text included memorized content.

Memorizing training data. The privacy risks of data memorization, for example the ability to extract sensitive data such as valid phone numbers and IRC usernames, are highlighted by [Carlini et al. \(2020\)](#). While their paper finds 604 samples that GPT-2 emitted from its training set, we show that *over 1%* of the data most models emit is memorized training data. In computer vision, memorization of training data has been studied from various angles for both discriminative and generative models (e.g. [Arpit et al., 2017](#); [Webster et al., 2019](#); [Feldman and Zhang, 2020](#); [Stephenson et al., 2021](#); [Teterwak et al., 2021](#)).

Duplicate text in training data. The Book Corpus ([Zhu et al., 2015](#)), which was used to train popular models such as BERT, has a substantial amount of exact-duplicate documents according to [Bandy and Vincent \(2021\)](#). [Allamanis \(2019\)](#) shows that duplicate examples in code datasets cause worsened performance on code understanding tasks.

3 Language Modeling Datasets

We analyze the presence of duplicate text in four datasets of varying sizes that have been used for training natural language generation systems, producing general-purpose pre-trained models, and for language model benchmarking. While this paper restricts itself to English datasets, we expect that non-English datasets suffer from similar issues and could likewise benefit from de-duplication.

Wikipedia (Wiki-40B) consists of multi-lingual cleaned Wikipedia text ([Guo et al., 2020](#)). We take the English portion, which contains 2.9M Wikipedia pages with an average length of 768 BPE tokens. The dataset creators do not indicate any deduplication was performed aside from removing redirect-pages (e.g., “sunflower” to “Helianthus”).

One-Billion Word benchmark (LM1B) contains 30M sentences of news commentary ([Chelba et al., 2013](#)). Unlike the other datasets we analyze, LM1B’s examples are one sentence long rather than multi-sentence documents. The average example length is 32 BPE tokens. While this dataset is extremely standard for benchmarking language models, [Radford et al. \(2019, Sec 4\)](#) note it has 13.2% overlap of the test set with the train set.

Colossal Cleaned Common Crawl (C4) is made up of 360M web documents, with an average length of 486 BPE tokens ([Raffel et al., 2020](#)). C4

was introduced as a pre-training dataset for T5, a set of encoder-decoder models which have been widely used in fine-tuned downstream tasks. The dataset was previously deduplicated in a more sophisticated process than the prior two datasets. Each paragraph was hashed and paragraphs resulting in hash collisions were removed. This was followed by a pass that removed placeholder text, code, and prohibited words. See [Dodge et al. \(2021a\)](#) for a detailed breakdown of the source text in C4.

RealNews is a subset of the Common Crawl consisting of articles from news domains ([Zellers et al., 2019](#)). It contains 31M documents with average length 793 BPE tokens. RealNews was deduplicated by inserting a hash of the first 100 characters of each document into a bloom filter ([Bloom, 1970](#)) and then excluding any document which resulted in a hash collision. Like C4, examples with duplicate URLs were excluded.

4 Methods for Identifying Duplicates

The simplest technique to find duplicate examples would be to perform exact string matching between all example pairs, but as we will show, this is insufficient. We introduce two complementary methods for performing deduplication. First, using a suffix array ([Manber and Myers, 1993](#)), we remove duplicate substrings from the dataset if they occur verbatim in more than one example. Second, we use MinHash ([Broder, 1997](#)), an efficient algorithm for estimating the n -gram similarity between all pairs of examples in a corpus, to remove entire examples from the dataset if they have high n -gram overlap with any other example.

We consider a dataset $D = \{x_i\}_{i=1}^N$ as a collection of *examples* x_i . Each of these examples is itself a sequence of *tokens*: $x_i = [x_i^1, x_i^2, \dots, x_i^{s_i}]$.

4.1 Exact Substring Duplication

Due to the diversity of possibilities in human language, it is rare for the same idea to be expressed identically in multiple documents unless one expression is derived from the other, or both are quoting from a shared source. This observation motivates deduplicating exact substrings. We call our approach EXACTSUBSTR. When two examples x_i and x_j share a sufficiently long substring (that is, a substring for which $x_i^{a..a+k} = x_j^{b..b+k}$), that substring is removed from one of them. Based on statistical analyses (§B), we select $k = 50$ tokens as the minimum matching substring length.